MAKRAM ALMOGHRABI

Full Stack Software Engineer | MSc Computer Science Engineer (BME)

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PROFESSIONAL SUMMARY

Analytical Software Engineer currently pursuing an MSc in Computer Science Engineering at the Budapest University of Technology and Economics (BME). Over 4 years of experience in full-stack development, specializing in React.js, Firebase, and secure system architecture. Proven expertise in bridging the gap between hardware (IoT/Sensors) and software (Web/Mobile). **All MSc coursework completed; available for immediate full-time employment.**

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), PHP, Flutter (Dart), HTML5, CSS3/Bootstrap.
- **Frameworks/Libraries:** React.js, Jquery, AJAX.
- **Cloud & Database:** NoSQL (Firebase), MySQL, ThingSpeak (Cloud IoT).
- **Networking & Security:** CCNA Certified (Routing, Switching, Wireless), Microsoft Security Fundamentals (Master Level), SSL, OAuth, SQL Injection Prevention.
- **Tools:** Git/GitHub, MATLAB (Data Processing), MQTT (IoT Protocol), ESP8266.
- **CMS/No-Code:** WordPress, Shopify, Squarespace.

EDUCATION

Budapest University of Technology and Economics (BME) | 2024 – Feb 2027 *MSc in Computer Science Engineering (Specialization: Software Engineering & Smart Cities)*

- **Coursework:** Advanced software architecture, Time Series Analysis for urban environments, and Cloud-based IoT implementation.

Lebanese International University | 2019 – 2022 *Bachelor of Science in Computer Science*

- **Academic Excellence:** Graduated with a **3.56 GPA**.

DETAILED ENGINEERING PROJECTS

- **Smart City Agent (Full Stack/AI):**

    - Developed a React-based dashboard integrating real-time Budapest city APIs (BKK/Public Services) with an AI chat interface.
    - Implemented specialized agents to parse natural language queries and visualize live transport routes and service locations on interactive maps.
- **BeatSync HRV Analysis (Mobile & HealthTech):**
    - Engineered a mobile application using Flutter to process Heart Rate Variability (HRV) data.
    - Integrated hardware sensors with mobile software for real-time health monitoring and data visualization.
- **Collaborative Task Manager (Complex UI):**
    - Architected a project management tool featuring custom drag-and-drop team workflows.
    - Developed a unique feature allowing users to pin comments to specific pixel coordinates on uploaded design assets for precise feedback.
- **Cloud-Based Weather Monitoring (IoT):**
    - Implemented a dual-approach system: A cloud system using ThingSpeak/MATLAB to process regional Hungarian weather data and a hardware-based system using ESP8266/AHT10 sensors via MQTT.
- **Computer Vision Meter Reader:**
    - Designed a utility tool that utilizes camera-based image recognition to automate water utility meter readings.

PROFESSIONAL EXPERIENCE

Full Stack Developer & Web Designer | Carrot Media | 2022 – 2025

- Architected and deployed custom web applications and e-commerce platforms using React.js and Firebase.
- Managed end-to-end development of digital card systems, streamlining professional networking for agency clients.
- Partnered with leading agencies (The Social Bang, Brainy Media) to deliver high-performance digital solutions under tight deadlines.

Full Stack Developer | Freelance | 2023 – Current

- Providing scalable website development and UI/UX design for international clients.
- Implementing secure backend infrastructures using PHP and MySQL to ensure data integrity.

Full Stack Developer | AWK Development (Part-time) | 2021 – 2022

- Developed and optimized backend systems using PHP, CSS, and JavaScript.
- Integrated AJAX calls and MySQL databases to enhance application responsiveness.

CERTIFICATIONS

- **CCNA Routing & Switching:** Scaling Networks (Cisco).
- **CCNAv7:** Switching, Routing, and Wireless Essentials (Cisco).
- **Microsoft Security Fundamentals:** Master Level Certification.
- **Digital Marketing Fundamentals:** Maharat min Google.
- **Cloud Computing:** IEEE LIU Student Branch Workshop.